MUSB
RESEARCH

PLASTISHEILD™

A Human-Origin Microplastic Mitigation
Probiotic & Postbiotic Platform

Technology Overview

PlastiSheild™ is a proprietary, human-origin probiotic and postbiotic platform engineered to address one of the fastest emerging environmental health concerns: **microplastic exposure and intestinal burden.**
This strain has been:

- Fully purified and phenotypically characterized
- Identified as Gram-positive and catalase-negative
- Screened to be free of major antibiotic resistance and pathogenic genes
- Whole genome sequenced for identity, safety, and stability
- Deposited with ATCC for traceability and reproducibility
- Scaled under GMP-compliant manufacturing conditions

PlastiSheild™ is positioned as a **first-in-class microbiome-based environmental defense technology**, targeting intestinal microplastic interactions and associated cellular stress responses.

The Emerging Environmental Health Challenge

Microplastics are now detected in:

- Drinking water
- Airborne particulates
- Food supply chains
- Human biological tissues

Chronic low-level exposure has been associated in emerging literature with:

- Intestinal inflammation
- Epithelial barrier disruption
- Oxidative stress
- Immune activation
- Systemic metabolic disturbances

Despite growing global awareness, there are currently no microbiome-targeted, biologically engineered interventions designed specifically to mitigate intestinal microplastic burden.

Innovation Rationale

PlastiSheild™ was developed to:

- Interact with microplastic particles within the gut lumen
- Reduce cellular interaction and stress responses in intestinal models
- Support epithelial barrier resilience
- Provide a preventive strategy within the emerging “exposome health” framework

This platform introduces a microbiome-mediated environmental resilience strategy, addressing both physical and biological aspects of exposure.

TECHNOLOGY CHARACTERIZATION

Strain Origin & Purification

PlastiSheild™ is derived from a human-origin microbial isolate, purified through rigorous microbiological and molecular selection processes.
Key characteristics:

- Single-strain isolation
- Gram-positive classification
- Stable genetic identity
- Catalase-negative
- Core probiotic functionality validated

Identity confirmation was achieved through whole genome sequencing and comparative genomic analysis.

Genomic & Safety Profiling

Whole genome sequencing confirms:

- Absence of significant virulence-associated genes
- No major transferable antibiotic resistance determinants
- Stable chromosomal architecture
- Absence of plasmid-mediated pathogenic markers

Genomic evaluation aligns with:

- FDA GRAS framework expectations
- EFSA QPS principles
- International probiotic safety standards

Functional Characterization

Preclinical functional screening included:

- Adhesion and epithelial compatibility assays
- Microplastic surface interaction analysis
- Intestinal cell viability and stress marker evaluation
- Barrier integrity testing

The strain demonstrates:

- Stable growth characteristics
- Controlled fermentation performance
- Favorable safety profile
- Functional resilience under physiological conditions

ATCC Deposition

Deposition in ATCC ensures:

- Global scientific traceability
- Independent verification capability
- Intellectual property protection
- Regulatory transparency

This strengthens sponsor confidence and long-term commercial viability.

MECHANISTIC EVIDENCE

Intestinal Cell Model Findings

Controlled in vitro studies using human intestinal epithelial cell models demonstrate:

- Reduction in microplastic-associated cellular stress markers
- Decreased epithelial oxidative stress responses
- Improved tight junction protein expression
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Compared to control conditions, PlastiSheild™ shows protective biological interaction under microplastic exposure models.

Proposed Mechanisms Under Investigation

Mechanistic pathways currently under evaluation include:

- Physical binding or sequestration of microplastic particles in the gut lumen
- Surface interaction modulation reducing epithelial contact
- Modulation of inflammatory cytokine signaling pathways
- Enhancement of tight junction proteins
- Postbiotic-mediated epithelial resilience

Further mechanistic validation studies are ongoing.

Postbiotic Activity

Heat-treated and cell-derived fractions of PlastiSheild™ demonstrate:

- Retained bioactivity in stress-reduction assays
- Barrier-supportive activity independent of live-cell replication
- Enhanced formulation flexibility

This supports development of:

- Live probiotic formulations
- Postbiotic-based products
- Hybrid synbiotic systems

DEVELOPMENT & COMPLIANCE

GMP Manufacturing

PlastiSheild™ is currently:

- Produced in GMP-certified fermentation facilities
- Manufactured under validated, controlled processes
- Subjected to batch identity and purity testing
- Stability evaluated under standard storage conditions

Quality control parameters include:

- Identity verification
- Microbial purity testing
- Potency assessment
- Contaminant screening
- Stability profiling

GRAS Preparation

Preparation for GRAS includes:

- Comprehensive genomic safety documentation
- Toxicology data review
- Manufacturing process controls
- Expert panel evaluation

The GRAS dossier will support positioning as a safe ingredient for human consumption.

Regulatory Positioning Strategy

Potential regulatory categories include:

- Dietary supplement ingredient
- Environmental wellness formulation
- Gut barrier support product
- Preventive exposome-targeted health product

Claims strategy will be structured around:

- Gut health support
- Microbial purity testing
- Barrier resilience
- Support against environmental stressors

Planned Clinical Endpoints

Future human studies may assess:

- Emerging microplastic biomarkers
- Intestinal permeability markers (e.g., zonulin, LPS)
- Inflammatory cytokine profiles
- Oxidative stress indicators
- Microbiome compositional changes
- Gastrointestinal symptom indices

Clinical studies will be randomized, placebo-controlled where appropriate.

COMMERCIAL PATHWAY

Target Markets

PlastiSheild™ is positioned for:

- Environmental health supplements
- Gut barrier protection formulations
- Exposome-targeted preventive wellness
- Corporate wellness programs
- Medical nutrition innovation

Product Formats

Potential formats include:

- Potential formats include:
- Synbiotic blends
- Functional beverages
- Preventive wellness kits
- Combination microbiome formulas

Co-Development Opportunities

MusB Research offers flexible partnership models:

- Licensing agreements
- Exclusive regional distribution
- Joint clinical validation programs
- Co-branded environmental health initiatives
- Strategic research partnerships

Strategic Value Proposition

PlastiSheild™ offers:

- First-mover advantage in exposome microbiome science
- Genomically transparent, human-origin strain
- Mechanistic validation in intestinal models
- Regulatory-ready development pathway
- GMP manufacturing scalability
- Strong differentiation in environmental wellness category

PLASTISHEILD™

PlastiSheild™ represents a pioneering step in microbiome-based environmental resilience. As global exposure to microplastics increases, there is a growing need for scientifically grounded, biologically rational solutions that address intestinal burden and barrier health.
Through rigorous genomic characterization, mechanistic validation, GMP-compliant manufacturing, and a clear regulatory roadmap, MusB Research offers sponsors a scalable and defensible innovation platform.

PlastiSheild™ is not merely a probiotic—it is a forward-looking solution at the intersection of microbiome science, environmental health, and translational innovation.

We invite strategic partners to collaborate in shaping the next generation of exposome-targeted health solutions.